

Carbonate alkalinity and dietary protein levels affected growth performance, intestinal immune responses and intestinal microflora in Songpu mirror carp (*Cyprinus carpio* Songpu)

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ABSTRACT

An eight week 3×2 two-factorial experiment was conducted to investigate the interaction between carbonate alkalinity levels at 0 and 15 mmol/L, and dietary protein levels at 310, 340 and 370 g/kg on the growth rate, digestive ability and the intestinal immunity, physical barrier function and microflora of Songpu mirror carp (*Cyprinus carpio* songpu). Fish at an initial average weight of 17.86 ± 0.19 g were randomly allotted to 18 aquariums each with 30 fish. Results revealed that weight gain rate (WGR) significantly decreased as carbonate alkalinity levels increased ($p < 0.05$), whereas the feed conversion ratio (FCR) of fish significantly increased ($p < 0.05$). Amylase activity in the intestine significantly decreased with increasing carbonate alkalinity levels ($p < 0.05$). A carbonate alkalinity of 15 mmol/L significantly reduced the superoxide dismutase (SOD) and total antioxidant capacity (T-AOC) activities, elevated the malondialdehyde (MDA) contents in the intestine and hepatopancreas and decreased the contents of Complement 3 (C3), Complement (C4) and Ig M ($p < 0.05$). The relative expressions of occludin, claudin3, claudin7, myosin light chain kinase (MLCK) and interleukin 10 (IL-10) in the intestine were significantly down-regulated with increasing carbonate alkalinity levels ($p < 0.05$), whereas the relative expressions of interleukin 6 (IL-6), interleukin 8 (IL-8) and tumor necrosis factor (TNF- α) in the intestine were significantly up-regulated ($p < 0.05$). The *Proteobacteria* abundance significantly increased with increasing carbonate alkalinity level ($p < 0.05$). The WGR significantly decreased as dietary protein levels increased ($p < 0.05$). The MDA contents in intestine and hepatopancreas also showed a significant rise with increasing dietary protein levels ($p < 0.05$). Increments in dietary protein levels significantly reduced the C3 and C4 contents of fish intestine ($p < 0.05$). Fish in the 370 g/kg diet protein level group showed a significantly lower relative expression of occludin and TNF- α ($p < 0.05$). There was an effect on the C3 content, the relative expression of occludin, the abundance of *Firmicutes* and *Firmicutes*: *Bacteroidetes* ratio between carbonate alkalinity and dietary proteins over time ($p < 0.05$).

In summary, under the long-term carbonate alkalinity stress of 15 mmol/L, dietary protein level for Songpu mirror carp should be controlled below 310 g/kg, which could effectively weaken the oxidative damage produced by carbonate alkalinity stress and further strengthen the growth and intestinal immune response of fish.

1. Introduction

In China, low-lying saline-alkali water covers an area of about 4.6×10^{11} m², which is widely distributed in 19 provinces, cities, and autonomous regions in the Northwest China, North and Northeast China. In these areas, carbonate alkalinity has been an important factor affecting the growth performance of fish (Liu et al., 2016a, 2016b). A

previous study showed that the growth of *Chalcalburnus chalcoides* was depressed when fed at a carbonate alkalinity of 40.78 mmol/L for 30 d (Jiang et al., 2008). The influencing mechanism of carbonate alkalinity on fish growth embraces many aspects. Among these, abnormal amino acid metabolism is one of the more critical factors (Zhao et al., 2013). Studies from this laboratory showed that chronic carbonate alkalinity stress at 15 and 30 mmol/L can significantly inhibit the

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growth and weaken the antioxidant capacity and ammonia metabolism of Songpu mirror carp (*Cyprinus carpio* Songpu) (Fan et al., 2020a, 2020b). In addition, Dang et al. (2013) verified on examination of the microstructure of the gill of juvenile *Barbus capito*, that the epithelial cells of the gill slice appeared swollen and necrotic and the volume of the secreting chlorine cells expanded with the increasing carbonate alkalinity, which restricted the growth of juvenile *Barbus capito*. A study of Nile tilapia *Oreochromis niloticus* revealed that there was an increase in the expression of the Rh gene and protein involved in the process of ammonia transport to address the elevated blood ammonia levels under acute alkalinity stress or chronic alkalinity stress (Tu et al., 2019). Sun et al. (2018) reported that according to high throughput metabolomics, 23 metabolic pathways of seven differential metabolites of crucian carp (*Carassius auratus*) were significantly affected and alkalinity exposure resulted in widespread change in metabolic profiles in the plasma with disruptions to the phenylalanine, glycine, serine, threonine, pyruvate and tyrosine metabolic pathways. It is noteworthy that amino acids in vivo are provided with dietary proteins for fish. Therefore, it is essential for fish to explore the protein requirement under the circumstance of the high-concentration carbonate alkalinity.

Protein is an essential nutrient for the growth, development and normal life activities of fish and is strongly associated with their immune function (Sanz et al., 2000). Protein insufficiency in diets can exert negative influence on growth and immunity (Oliva Teles et al., 2020; Sun et al., 2016), whereas dietary protein excess may bring about an ammonia metabolic burden, resulting in poor growth and the decrease in immunity. The external elements that affect the optimum protein level of fish include feed raw material sources, feed processing technology, feeding ratio, feeding frequency, feeding time, stocking density and environmental factors such as temperature, ammonia nitrogen, alkalinity, pH and salinity (Zehra and Khan, 2012). Environmental factors are the most unstable and uncontrollable and play a significant role in fish growth and immune function development (Sun et al., 2016), but in other previous studies of protein nutrition in fish, water temperature and salinity were more significant (Degani et al., 1989; Zeitoun et al., 2011; El-Sayed et al., 2003). Up to now, information on the optimal dietary protein requirement of fish in condition of the high-concentration carbonate alkalinity is not available.

Songpu mirror carp (*Cyprinus carpio* Songpu) has been successfully bred through multi-character complex population selection combined with microsatellite and electronic marking technology, which has the advantages of high survival rate, strong disease and cold resistance and an improved saline-alkaline tolerance. To date, *C. carpio* Songpu have been bred on a large scale in most areas of China (Hu et al., 2017; Shi et al., 2016). At present, there were a number of univariate studies on the influence of carbonate alkalinity or protein level on Songpu mirror carp (Xu et al., 2014; Fan et al., 2020a, 2020b), but no studies have been conducted to evaluate the interaction of carbonate alkalinity and protein levels on these fish.

The present study was carried out over an eight week period to evaluate the interaction effects of prolonged exposure to different carbonate alkalinity levels and different dietary protein levels on growth rates, digestive ability, non-specific immunity and intestinal microflora of Songpu mirror carp. The results will provide scientific guidance for helping Songpu mirror carp resist carbonate alkalinity stress in its local aquatic environment.

2. Materials and methods

2.1. Ethics statement

All Animal procedures in this study were conducted according to the guidelines for the care and use of laboratory animals of Heilongjiang River Fisheries Research Institute (CAFS). The studies in animals were reviewed and approved by the Committee for the Welfare and Ethics of Laboratory Animals of Heilongjiang River Fisheries Research Institute

(CAFS).

2.2. Experimental diet formulation and preparation

Based on Formula feed for common carp (GB/T 36782–2018) (2019) and NRC (2011) guidelines, the approximate composition analysis of the experimental diets are presented in Table 1. Soybean oil, fish oil, phospholipid and flour were dietary lipid and carbohydrate sources, respectively. Three diets were formulated to yield 30 g/kg increments of protein from 310 to 370 g/kg diet. The flour was equally shrinked as the dietary protein CP levels increased to ensure that the dietary available energy was alike for all formulations. The ingredients of the diets were thoroughly mixed and soybean oil, fish oil, phospholipids and distilled water were added until a smooth dough was obtained. All the diets were granulated into 1.5 mm pellets using a pelletizer (GYJ-250B, Dashiqiao Bao's Feed Machinery Factory). The resulting pellets were air dried and stored at -20°C until used.

A representative diet sample was taken for a composition analysis according to the standard procedure of the AOAC (1995). Moisture was determined by oven drying to a constant weight at 105°C . Crude protein was analyzed by the Kjeldahl method after acid digestion. Crude lipids were determined by using the other extraction method with a Soxtec system. Crude ash was determined after combustion to a constant weight at 550°C .

2.3. Fish management and feeding

The experiment was conducted in an indoor aquarium with a controlled water circulation system, a water volume of 200 L and an oxygen auto-supplemented system. Juvenile songpu mirror carp were obtained from Hulan Experimental Station of Heilongjiang Fisheries Research Institute (Harbin, China), and acclimated to the laboratory conditions for two weeks. During acclimation, fish were fed three times a day with a commercial diet containing 320 g/kg crude protein (Tong Wei Co. Ltd.). After acclimation, thirty fish with an average initial

Table 1
Composition and nutrient levels of the basal diet (DM basis) g/kg.

Ingredients	310	340	370
Soybean meal	480	580	660
Flour	310	200	155
Fish meal	100	100	100
Soybean oil	30	30	18
Phospholipid	10	10	10
Fish oil	10	10	10
CMCC-Na	10	10	10
Choline chloride	5	5	5
Ca(H ₂ PO ₄) ₂	25	25	25
Vitamin premix ¹	3	3	3
Mineral premix ²	2	2	2
Cellulose	5.1	21.5	0.6
Lysine	5	0.5	0
Methionine	2.2	1.8	1.4
Threonine	2.5	1.2	0
Total	1000	1000	1000
Nutrient levels			
Crude protein /CP ³	311.5	341.2	37,006
Ether extract/ EE	65.8	65.9	54.6
Crude Ash	70.8	76.6	81.5
Met	7.1	7.1	7.1
Lys	22.1	22.2	23.8
Thr	15.2	15.0	14.9

¹ The vitamin premix provided the following per kg of the diet: VA 8000 IU, VC 500 mg, VD₃ 3000 IU, VE 60 mg, VK₃ 5 mg, VB₂ 30 mg, VB₆ 15 mg, VB₁₂ 0.5 mg, choline chloride 5000 mg, nicotinic acid 175 mg, D-biotin 2.5 mg, inositol 1000 mg, folic acid 5 mg, pantothenic acid 50 mg.

² The mineral premix provided the following per kg of the diet: Zn 25 mg, Cu 3 mg, Fe 25 mg, Mn 15 mg, I 0.6 mg, Co 0.1 mg, Se 0.4 mg.

³ CP and EE are measured values, while the others are calculated values.

weight of 17.86 ± 0.19 g were randomly assigned to each of 18 experimental aquariums. During the trial, the water temperature ranged from 22 to 24 °C. The water dissolved oxygen was over five mg/L. 30% of water was changed every two days and the fish tank was cleaned regularly to maintain good water quality. And then, the water was adjusted to the appropriate concentration with the preconfigured mother liquor. The fish were hand-fed to apparent satiety three times daily at 08:00, 12:00, and 17:00 for the eight weeks of the trial.

2.4. Carbonate alkalinity design

Based on the previous study in our laboratory (Fan et al., 2020a, 2020b), a freshwater control group with 0 mmol/L carbonate alkalinity and 15 mmol/L carbonate alkalinity along with three parallel groups were designed for each protein treatment group. Analytically pure sodium bicarbonate, with a product purity $\geq 99.5\%$ (Tianjin Zhiyuan Chemical Reagent Co., Ltd) was used to prepare the corresponding carbonate alkalinity and 0.02 mmol/L HCl was used to calibrate it. Mixed methyl orange-aniline blue and phenolphthalein were used as indicators and the measured carbonate alkalinity and pH are detailed in Table 2.

2.5. Sample collection and analysis

At the end of the feeding trial, the fish in each aquarium were counted, and batch weighed after 24 h food deprivation. Three fish in each aquarium were weighed and measured individually to calculate the hepatosomatic index (HSI), condition factor (CF) and viscerosomatic index (VSI). They were then dissected on an ice box to acquire the hepatopancreas and the intestine for enzyme activity test and these were frozen at -20 °C before analysis. Another three fish from each aquarium were dissected on an ice box to obtain the intestine for detection of intestinal microflora, which were stored in a freezer tube, by being first deep frozen in liquid nitrogen and then saved at -80 °C before detection.

2.6. Determination of enzyme activity

The hepatopancreas and the intestine for enzyme activity test were homogenized in nine volumes of ice cold physiological saline (0.9% w/v NaCl) and centrifuged at 3500 rpm for 25 min at 4 °C (Thermo Fisher Scientific ST 16R centrifuge), with the resulting supernatants aliquoted and kept at -20 °C until the next analysis. The activities of superoxide dismutase (SOD), and catalase (CAT) in the hepatopancreas and the intestine were measured using kits which were produced by Nanjing Jiancheng Bioengineering Institute (Cat. No A001–1, A006–1, A007–1). The activities of amylase, lipase in the intestine were measured using kits that were produced by Nanjing Jiancheng Bioengineering Institute (Cat. No C016–1, A054–1). Protease activity was assayed by the folin-reagent method. One unit of protease activity was defined as one μ g tyrosine released one gram tissue in one min at 37 °C (Liang et al., 2018). Protein contents in the hepatopancreas and the intestine were measured based on coomassie blue staining using the kit which was produced by Nanjing Jiancheng Bioengineering Institute (Cat. No A045–2-2).

Table 2

Designed and measured carbonate alkalinity. **N** = 6; means $\pm$ SD; mmol/L.

Caobonate alkalinity and pH	Groups	
	0 mmol/L	15 mmol/L
Designed caobonate alkalinity	0.00	15.0
Measured caobonate alkalinity	0.00	14.9 $\pm$ 0.1
pH	6.95 $\pm$ 0.16	8.27 $\pm$ 0.16

2.7. Quantitative real-time PCR (qPCR)

The total RNA was isolated with RNAiso Plus according to the manufacturer's instructions. The purity and quantitative OD260/280 of the isolated RNA were quantified by spectrophotometry at 260 and 280 nm respectively, whereas the quality of the RNA was verified by agarose gel electrophoresis. The first-strand cDNA was synthesized with the TaKaRa PrimeScript™ RT reagent Kit with gDNA Eraser (Perfect Real Time) (Code No: RR047A) (Dalian Takara Company). The qPCR was conducted with the cDNA template and the TaKaRa SYBR® Primix Ex Taq™ (Tli RNaseH Plus)(Code No: RR420A) (Dalian Takara Company) following the manufacturer's protocol. The primers are listed in Table 3. The β -actin was selected as the internal standard because of its stable expression in different tissues over time. The $2^{-\Delta\Delta Ct}$ method was used to determine the relative expression levels of each gene.

2.8. Intestinal microflora experiment procedure

Intestine samples from three fish in each aquarium were combined for intestinal microbiota analysis. DNA amplification and sequencing were conducted by Shanghai Majorbio Bio-pharm Technology Co., Ltd. according to the manufacturer's protocol for extraction of microbial DNA from intestinal metabolite samples using the E.Z.N.A stool DNA Kit (Omega Bio-Tek). Microbial genomic DNA was extracted from each fecalsample (0.2 g) using the cetyltrimethylammonium bromide/sodium dodecyl sulfate method. DNA concentration and purity were evaluated on 1% agarose gels. The extracted DNA was diluted to a concentration of 1 ng/ μ L using sterile water. After quantification, qualification, mixing, and purification of the PCR products, the V4 region of the 16S rRNA was sequenced using an Illumina MiSeq platform in accordance with the standard protocol of the manufacturer. Raw data were assembled using the FLASH analysis tool and filtered with the Quantitative Insights into Microbial Ecology (QIIME) software package (<http://qiime.org/>). Chimera sequences were removed with the UCHIME algorithm (http://drive5.com/usearch/manual/uchime_algo.html) to obtain effective tags. Sequences with $\geq 97\%$ similarity, as analyzed with Uparse software (<http://drive5.com/uparse/>), were

Table 3

Primers used for quantitative RT-PCR (qPCR).

Gene name	Primer sequence(5'-3')	Gene serial number
occludin	F:ATCGGTTTCAGTACAATCAGG R:GACAATGAAGCCCATTAACAA	KF975606
ZO-1 ¹	F:GCCTGCCTACACTCAACCACAAC R:CTGCTTCGGCTGGAGGAGGAG	KY290394.1
claudin3	F:GCACCAACTGTATCGAGGATG R:GGTTGTAGAAGTCCCGAATGG	JQ767157
claudin7	F:CTTCTATAACCCCTTCACACCAG R:ACATGCCTCCACCCATTATG	JQ767155
claudin11	F:TCGGAAGTGAACCGAAGAGC R:GAAGCCAAAGGACATCAAGC	JQ767158
MLCK ²	F:AGCAGTGTGGGATCAACCT R:CTCCAGCAGGGTCATGATGAG	XM_019076433.1
IL-1 β ³	F:AACTTCACACTTGAGGAT R: GACAGAACAATAACAACAAC	KC008576
IL-6 ⁴	F: TAGGTTAATGAGCAAGAGGA R: AGAGACTGTTGATACTGGAA	AY102633.1
IL-8 ⁵	F:AAACTGAGAGTCGACGCATTG R:TTTTCATGACCTTCTTAACCCAG	EU011243.1
IL-10 ⁶	F:GCCAGCATAAAGAACTCG R:CCAAATACTGCTCGATGT	JX524550.1
TNF- α ⁷	F:AAGTCTCAGAACAATCAGGAA R: TGCCTTGGAAAGTGACATT	AJ311800
β -actin	F:GATCGGCAATGAGCGTTTCC R: ACGGTGTTGGCATAACAGGTC	M24113.1

Note: 1. ZO-1: Zonula occludin 1; 2. MLCK: Myosin Light Chain Kinase; 3. IL-1 β : Interleukin-1 β ; 4. IL-6: Interleukin-6; 5. IL-8: Interleukin-8; 6. IL-10: Interleukin-10; 7. TNF: Tumor Necrosis Factor; And F stands for Forward; R stands for Reverse.

assigned to the same operational taxonomic unit (OTU) (Zhang et al., 2017). In order to compute alpha diversity to analyze the complexity of species diversity for a sample, observed species were applied. Beta diversity of both the weighted and unweighted UniFrac metrics, which consider the presence/absence of taxa between sample pairs, was calculated using QIIME software. Cluster analysis was preceded by principal component analysis. Principal coordinate analysis (PCoA) was performed to visualize principal coordinates from complex, multidimensional data. The unweighted pairgroup method with arithmetic means (UPGMA) hierarchical clustering was to interpret the distance matrix using average linkages. All original sequences were downloaded from the National Center for Biotechnology Information database (<https://www.ncbi.nlm.nih.gov/pubmed/>).

2.9. Data processing and analysis

Weight growth rate (WGR, %) = $100\% \times (W_t - W_0) / W_0$.

Feed conversion ratio (FCR) = $F / (W_t - W_0)$.

Hepatosomatic index (HIS, %) = $100\% \times W_h / W_t$.

Condition factor (CF, g/cm³) = $100\% \times W_t / L_t^3$.

Viscerosomatic index (VSI, %) = $100\% \times W_v / W_t$.

Where N_t and N_0 are the final and initial body number, W_t and W_0 are the final and initial body weight (g), t is time of rearing (days), L is fish body length (cm), F is total feed intake (g), W_g is liver weight (g), W_v is viscera weight (g).

Results were subjected to two-way analysis of variance (ANOVA), followed by a Duncan's test to delineate significance among fish groups using SPSS 22.0 software (SPSS Inc., Chicago, IL, USA). For all analyzes, significant levels were set at $p < 0.05$ unless otherwise stated. All data are represented as means $\pm$ SD.

3. Results and analysis

3.1. Growth parameters, feed utilization and morphological index

The results on the effects of carbonate alkalinity and dietary protein level on growth performance, feed utilization and morphological index of Songpu mirror carp are presented in Table 4 and Table 5.

Growth performance was significantly affected by carbonate alkalinity and dietary protein level ($p < 0.05$). Fish fed the diet containing 370 g/kg protein at the carbonate alkalinity of 15 mmol/L had the lowest mean final weight and WGR. Final weight and WGR of fish were significantly increased by increasing dietary protein from 310 to 470 g/kg at the same carbonate alkalinity level ($p < 0.05$) and were also significantly increased with increasing carbonate alkalinity levels from 0 to 15 mmol/L at the same dietary protein level ($p < 0.05$).

The feed conversion ratio (FCR) of fish fed at a carbonate alkalinity of 15 mmol/L was higher than that of fish fed at a carbonate alkalinity of 0 mmol/L ($p < 0.05$) and this was not affected by dietary protein levels ($p > 0.05$). Fish fed the diet containing 310 g/kg protein at a carbonate alkalinity of 0 mmol/L, with FCR of 1.35 ± 0.04 , exhibited the best growth performance and feed utilization of all trial fish. The poorest growth and feed utilization were found in the fish fed the diet containing 370 g/kg protein at a carbonate alkalinity of 15 mmol/L.

As seen in Table 5, condition factor (CF), hepatosomatic index (HIS) and viscerosomatic index (VSI) were not significantly affected by either the carbonate alkalinity or the dietary protein levels ($p > 0.05$). Regardless of the dietary protein levels, CF generally reduced as the carbonate alkalinity levels increased, whereas HIS tended to increase as the carbonate alkalinity levels increased, although no significant difference was observed ($p > 0.05$).

3.2. Digestive enzyme activity

Table 6 shows the changes in digestive enzyme activities of the trial fish. The carbonate alkalinity levels significantly influenced the amylase

Table 4

Growth performance and feed utilization of Songpu mirror carp under different treatments.

Carbonate alkalinity levels (mmol/L)	Protein levels (g/kg)	Initial weight (g)	Final weight (g)	Weight gain rate (WGR, %)	Feed conversion ratio (FCR)
0	310	18.00 $\pm$ 0.15	129.4 $\pm$ 7.6 ^a	618.8 $\pm$ 36.8 ^a	1.35 $\pm$ 0.04 ^c
	340	17.93 $\pm$ 0.13	120.8 $\pm$ 4 ^{bo}	574.1 $\pm$ 25.5 ^a	1.40 $\pm$ 0.05 ^c
	370	18.00 $\pm$ 0.06	112.4 $\pm$ 5.8 ^c	525.2 $\pm$ 32.7 ^{ab}	1.57 $\pm$ 0.08 ^b
15	310	17.60 $\pm$ 0.10	89.1 $\pm$ 1.8 ^d	405.5 $\pm$ 12.9 ^{bc}	1.81 $\pm$ 0.12 ^a
	340	17.80 $\pm$ 0.26	82.4 $\pm$ 3.6 ^d	362.9 $\pm$ 16.7 ^{cd}	1.89 $\pm$ 0.08 ^a
	370	17.83 $\pm$ 0.20	81.4 $\pm$ 1.3 ^d	356.8 $\pm$ 7.8 ^d	1.85 $\pm$ 0.11 ^a
Carbonate alkalinity	0	17.98	120.9 ^p	572.7 ^p	1.44 ^p
Protein levels	15	17.74	84.3 ^q	375.1 ^q	1.85 ^q
	310	17.80	109.3 ^x	512.1 ^x	1.58
	340	17.87	101.6 ^{xy}	468.5 ^{xy}	1.65
	370	17.92	96.9 ^y	441.0 ^y	1.71
P-value of two-way ANOVA					
Protein levels		0.78	0.04	0.04	0.35
Carbonate alkalinity		0.11	0.00	0.00	0.00
Carbonate alkalinity * Protein levels		0.69	0.58	0.60	0.40

Note: all the values are expressed as the mean values $\pm$ SD, and different letters in the same column denote a significant difference ($p < 0.05$) (the same as those shown in the following tables).

Table 5

Morphological index of Songpu mirror carp under different treatments.

Carbonate alkalinity levels (mmol/L)	Protein levels (g/kg)	Condition factor (CF, g/cm ³)	Hepatosomatic index (HIS, %)	Viscerosomatic index (VSI, %)
0	310	2.27 $\pm$ 0.01	3.30 $\pm$ 0.40	10.0 $\pm$ 0.7
	340	2.27 $\pm$ 0.07	2.97 $\pm$ 0.24	10.5 $\pm$ 0.8
	370	2.26 $\pm$ 0.07	3.36 $\pm$ 0.37	10.8 $\pm$ 0.4
15	310	2.14 $\pm$ 0.11	3.71 $\pm$ 0.38	10.0 $\pm$ 0.8
	340	2.17 $\pm$ 0.12	3.33 $\pm$ 0.27	11.4 $\pm$ 0.7
	370	2.26 $\pm$ 0.14	3.70 $\pm$ 0.21	9.9 $\pm$ 0.9
Carbonate alkalinity	0	2.27	3.21	10.4
	15	2.19	3.58	10.4
Protein levels	310	2.20	3.51	10.0
	340	2.22	3.15	10.9
	370	2.26	3.53	10.4
P-value of two-way ANOVA				
Protein levels		0.81	0.44	0.48
Carbonate alkalinity		0.35	0.19	0.97
Carbonate alkalinity * Protein levels		0.79	0.99	0.46

Note: all the values are expressed as the mean values $\pm$ SD, and different letters in the same column denote a significant difference ($p < 0.05$) (the same as those shown in the following tables).

and lipase activities ($p < 0.05$). The activities of amylase in fish significantly decreased with increasing carbonate alkalinity level from 0 to 15 mmol/L at the same dietary protein level ($p < 0.05$). Irrespective of the dietary protein levels, the activities of lipase significantly reduced as the carbonate alkalinity levels rose, whereas the activities of protease decreased, although no significant difference was observed ($p > 0.05$).

Dietary protein levels did not affect amylase, lipase, and protease

Table 6

Digestive enzyme activities in the intestine of Songpu mirror carp under different treatments.

Carbonate alkalinity levels (mmol/L)	Protein levels (g/kg)	Amylase (U/g prot)	Lipase (U/g prot)	Protease (U/g prot)
0	310	0.77 ± 0.07 ^a	15.4 ± 0.8 ^a	50.4 ± 1.8 ^a
	340	0.70 ± 0.01 ^{bc}	12.3 ± 0.1 ^b	47.7 ± 3.7 ^{ab}
	370	0.73 ± 0.02 ^{ab}	10.8 ± 1.3 ^{bc}	39.6 ± 5.8 ^c
15	310	0.58 ± 0.01 ^e	9.7 ± 1.7 ^c	42.1 ± 2.4 ^{bc}
	340	0.65 ± 0.01 ^{cd}	11.2 ± 1.1 ^{bc}	44.9 ± 1.4 ^{abc}
	370	0.61 ± 0.04 ^{de}	9.1 ± 1.4 ^c	38.8 ± 3.1 ^c
Carbonate alkalinity	0	0.73 ^p	12.8 ^p	45.9
	15	0.61 ^q	10.0 ^q	41.9
Protein levels	310	0.67	12.5	46.3
	340	0.68	11.8	46.3
	370	0.67	9.9	39.2
P-value of two-way ANOVA				
Protein levels		0.98	0.11	0.09
Carbonate alkalinity		0.01	0.01	0.17
Carbonate alkalinity * Protein levels		0.15	0.14	0.52

Note: all the values are expressed as the mean values ± SD, and different letters in the same column denote a significant difference ($p < 0.05$) (the same as those shown in the following tables).

activity ($p > 0.05$). Lipase and protease activities in the intestine of fish tended to reduce as dietary protein levels increased, but no significant difference was observed ($p > 0.05$).

3.3. Antioxidant capacity

According to the antioxidant enzyme activities of the intestine recorded in Table 7, carbonate alkalinity increases significantly decreased SOD and T-AOC activities in the intestine and significantly increased MDA contents ($p < 0.05$). Regardless of the dietary protein levels, a carbonate alkalinity of 15 mmol/L can significantly decline the SOD and T-AOC activities and elevate the MDA content ($p < 0.05$), compared with a carbonate alkalinity of 0 mmol/L. In addition, the

Table 7

Antioxidant enzyme activities in the intestine of Songpu mirror carp under different treatments.

Carbonate alkalinity levels (mmol/L)	Protein levels (g/kg)	Superoxide dismutase (SOD, U/mg prot)	Total antioxidant capacity (T-AOC, U/mg prot)	Malondialdehyde (MDA, nmol/mg prot)
0	310	82.0 ± 5.5 ^a	1.00 ± 0.03 ^a	0.39 ± 0.06 ^c
	340	78.0 ± 4.1 ^{ab}	0.86 ± 0.06 ^b	0.52 ± 0.12 ^{bc}
	370	75.2 ± 5.1 ^{ab}	0.99 ± 0.08 ^a	0.59 ± 0.09 ^b
15	310	70.3 ± 5.9 ^{bc}	0.64 ± 0.02 ^d	0.66 ± 0.12 ^{ab}
	340	64.0 ± 2.8 ^c	0.78 ± 0.04 ^{bc}	0.68 ± 0.04 ^{ab}
	370	69.5 ± 2.5 ^{bc}	0.68 ± 0.08 ^{cd}	0.81 ± 0.09 ^a
Carbonate alkalinity	0	78.4 ^p	0.95 ^p	0.50 ^p
	15	67.9 ^q	0.70 ^q	0.72 ^q
Protein levels	310	76.1	0.82	0.53
	340	70.1	0.82	0.60
	370	72.3	0.84	0.70
P-value of two-way ANOVA				
Protein levels		0.51	0.94	0.20
Carbonate alkalinity		0.01	0.00	0.01
Carbonate alkalinity * Protein levels		0.65	0.06	0.83

Note: all the values are expressed as the mean values ± SD, and different letters in the same column denote a significant difference ($p < 0.05$) (the same as those shown in the following tables).

activities of SOD in the intestine significantly decreased with increasing carbonate alkalinity level from 0 to 15 mmol/L at the dietary protein level of 310 and 340 g/kg ($p < 0.05$), the activities of T-AOC in the intestine significantly decreased with increasing carbonate alkalinity level from 0 to 15 mmol/L at the dietary protein level of 340 and 370 g/kg ($p < 0.05$) and the contents of MDA in the intestine significantly increased with increasing carbonate alkalinity levels from 0 to 15 mmol/L at a dietary protein level of 310 and 370 g/kg ($p < 0.05$).

Results in Table 8 showed significant effects of carbonate alkalinity on antioxidant enzyme activities in hepatopancreas, where increasing the carbonate alkalinity significantly decreased the T-AOC activities and significantly increased the MDA contents ($p < 0.05$). Irrespective of the dietary protein levels, the T-AOC activity was significantly decreased and the MDA contents were significantly increased by increasing the carbonate alkalinity level from 0 to 15 mmol/L ($P < 0.05$). The activities of SOD slightly decreased with increasing carbonate alkalinity level from 0 to 15 mmol/L at the dietary protein level of 310 and 340 g/kg, although there was no significant difference among the treatments ($p > 0.05$), the activities of T-AOC significantly decreased with increasing carbonate alkalinity level from 0 to 15 mmol/L at the dietary protein level of 340 and 370 g/kg ($p < 0.05$) and the contents of MDA in fish significantly increased with increasing carbonate alkalinity level from 0 to 15 mmol/L at the dietary protein levels of 310 and 370 g/kg ($p < 0.05$). Increasing the dietary protein levels had no statistically significant effect on the activities of SOD, T-AOC and MDA content in the intestine and hepatopancreas ($p > 0.05$).

3.4. Non-specific immunity

As shown in Table 8, increasing carbonate alkalinity levels significantly affected the contents of C3, C4 and Ig M in the hepatopancreas ($p < 0.05$). Compared with a carbonate alkalinity of 0 mmol/L, a carbonate alkalinity of 15 mmol/L can significantly decrease the contents of C3, C4 and Ig M in hepatopancreas ($p < 0.05$). At the same protein levels, the contents of C3, C4 and Ig M in the hepatopancreas significantly decreased with increasing carbonate alkalinity level from 0 to 15 mmol/L ($p < 0.05$).

A significant effect was also observed where increasing the dietary protein levels significantly decreased the C3 and C4 contents in the

Table 8

Non-specific immunity in hepatopancreas of Songpu mirror carp under different treatments.

Carbonate alkalinity levels (mmol/L)	Protein levels (g/kg)	C3 (mg/g prot)	C4 (mg/g prot)	Ig M (mg/g prot)
0	310	30.4 ± 1.1 ^a	62.8 ± 3.7 ^a	24.8 ± 2.4 ^a
	340	25.4 ± 1.5 ^b	56.4 ± 2.6 ^b	20.6 ± 2.5 ^b
	370	20.8 ± 1.3 ^c	53.5 ± 5.6 ^b	18.2 ± 1.5 ^{bc}
15	310	18.2 ± 0.7 ^{cd}	39.1 ± 3.4 ^c	15.0 ± 1.3 ^{cd}
	340	15.6 ± 0.7 ^{de}	44.0 ± 5.4 ^c	11.7 ± 1.8 ^d
	370	15.3 ± 2.6 ^e	41.3 ± 2.0 ^c	13.9 ± 2.9 ^d
Carbonate alkalinity	0	25.5 ^p	57.6 ^p	21.2 ^p
	15	16.4 ^q	41.5 ^q	13.5 ^q
Protein levels	310	24.3 ^x	50.9 ^x	19.9
	340	20.5 ^y	50.2 ^x	16.2
	370	18.0 ^z	47.4 ^y	16.1
P-value of two-way ANOVA				
Protein levels		0.00	0.01	0.66
Carbonate alkalinity		0.00	0.00	0.00
Carbonate alkalinity * Protein levels		0.00	0.09	0.30

Note: all the values are expressed as the mean values ± SD, and different letters in the same column denote a significant difference ($p < 0.05$) (the same as those shown in the following tables).

hepatopancreas ($p < 0.05$). At the same carbonate alkalinity levels, the contents of C3 and C4 in the hepatopancreas significantly decreased with increasing the dietary protein level from 310 to 370 g/kg ($p < 0.05$). Increasing the dietary protein levels had no statistically significant effect on Ig M in hepatopancreas ($p > 0.05$), but the contents of IgM tended to decrease with increasing dietary protein levels, although there was no statistically significant difference ($p > 0.05$).

It is worth noting that the C3 content was significantly affected by the interaction of dietary protein level and carbonate alkalinity levels ($p < 0.05$), where the lowest C3 content was found in fish fed the diet containing 370 g/kg protein at a carbonate alkalinity level of 15 mmol/L.

3.5. Relative expression of tight junction complexes (TJs) and myosin light chain kinase (MLCK)

As shown in Table 9, the relative expressions of occludin, claudin3, claudin7 and MLCK in the intestine were significantly influenced by carbonate alkalinity levels ($p > 0.05$) except for the relative expressions of zonula occludens 1 (ZO-1) and claudin11 ($p < 0.05$). The relative expressions of occludin, claudin3, claudin7 and MLCK in the intestine were significantly down-regulated with increasing carbonate alkalinity levels irrespective of the dietary protein levels. At the same dietary protein levels, a carbonate alkalinity level of 15 mmol/L could significantly down-regulate the relative expressions of occludin, claudin3, claudin7 and MLCK in the intestine compared to a carbonate alkalinity level of 0 mmol/L ($p < 0.05$). Increasing dietary protein levels only affected the relative expression of occludin, so fish fed the diets containing the protein level of 370 g/kg exhibit the lower relative expression of occludin compared to those of fish fed the diets containing the protein level of 310 and 340 g/kg ($p < 0.05$), irrespective of the carbonate alkalinity levels. The relative expression of occludin was significantly influenced by the interaction of carbonate alkalinity levels and dietary protein levels ($p < 0.05$).

As shown in Table 10, the relative expressions of IL-6, IL-8 and TNF- α in the intestine were significantly influenced by carbonate alkalinity levels ($p < 0.05$) except for the relative expressions of IL-1 β and IL-10 ($p > 0.05$). The relative expressions of IL-6, IL-8 and TNF- α in the intestine were significantly up-regulated with increasing carbonate alkalinity levels without regard to the dietary protein levels ($p < 0.05$), while the

Table 10

Relative expression of cytokines in the intestine of Songpu mirror carp under different treatments.

Carbonate alkalinity levels(mmol/L)	Protein levels(g/kg)	IL-1β	IL-6	IL-8	IL-10	TNF-α
0	310	1.49 ± 0.14	1.13 ± 0.11 ^b	0.65 ± 0.05 ^c	1.60 ± 0.31	1.24 ± 0.10 ^a
		1.56 ± 0.24	0.87 ± 0.05 ^b	0.69 ± 0.09 ^c	1.62 ± 0.31	1.03 ± 0.10 ^{ab}
		1.19 ± 0.24	0.81 ± 0.02 ^b	0.57 ± 0.03 ^c	1.32 ± 0.12	0.76 ± 0.13 ^{bc}
	340	1.66 ± 0.24	1.39 ± 0.30 ^a	1.17 ± 0.06 ^a	1.43 ± 0.07	0.94 ± 0.10 ^{bc}
		1.66 ± 0.20	1.38 ± 0.19 ^a	1.05 ± 0.16 ^{ab}	1.52 ± 0.10	0.94 ± 0.03 ^{bc}
		1.61 ± 0.06	1.10 ± 0.15 ^{ab}	0.81 ± 0.09 ^{bc}	1.26 ± 0.10	0.68 ± 0.03 ^c
	370	1.41 ± 0.06	0.94 ^q	0.64 ^q	1.51 ± 0.10	1.00 ^p
		1.64 ± 0.06	1.29 ^p	1.01 ^p	1.40 ± 0.10	0.84 ^q
		1.57 ± 0.06	1.26	0.91	1.52 ± 0.10	1.09 [×]
15	310	1.61 ± 0.06	1.13	0.87	1.57 ± 0.10	0.97 [×]
		1.40 ± 0.06	0.96	0.69	1.29 ± 0.10	0.72 ^y
		0.47	0.18	0.06	0.20	0.00
	340	0.13	0.01	0.00	0.41	0.03
		0.64	0.68	0.33	0.94	0.44
		0.47	0.18	0.06	0.20	0.00
	370	0.13	0.01	0.00	0.41	0.03
		0.64	0.68	0.33	0.94	0.44
		0.47	0.18	0.06	0.20	0.00
P-value of two-way ANOVA						
Protein levels						
Carbonate alkalinity						
Carbonate alkalinity * Protein levels						

Note: all the values are expressed as the mean values $\pm$ SD, and different letters in the same column denote a significant difference ($p < 0.05$) (the same as those shown in the following tables).

relative expressions of IL-10 in the intestine were down-regulated with increasing carbonate alkalinity levels ($p > 0.05$). Increasing dietary protein levels had a significant overall impact on the relative expression of TNF- α ($p < 0.05$). Fish fed the diets containing the protein level of 370 g/kg exhibit the lower relative expression of TNF- α compared to

Table 9

Relative expression of tight junction complexes (TJs) and Myosin light chain kinase (MLCK) in the intestine of Songpu mirror carp under different treatments.

Carbonate alkalinity levels (mmol/L)	Protein levels(g/ kg)	occludin	zonula occludens 1 (ZO-1)	claudin3	claudin7	claudin11	Myosin light chain kinase (MLCK)
0	310	1.15 ± 0.07 ^a	1.17 ± 0.07	0.66 ± 0.05 ^{ab}	0.74 ± 0.08 ^a	1.50 ± 0.29	1.34 ± 0.12 ^{ab}
	340	0.96 ± 0.08 ^b	1.11 ± 0.14	0.67 ± 0.02 ^a	0.63 ± 0.12 ^{ab}	1.29 ± 0.0.17	1.72 ± 0.22 ^a
	370	0.57 ± 0.02 ^d	0.84 ± 0.03	0.64 ± 0.11 ^{ab}	0.62 ± 0.05 ^{ab}	1.08 ± 0.23	1.33 ± 0.11 ^{ab}
15	310	0.76 ± 0.04 ^c	0.91 ± 0.18	0.40 ± 0.01 ^c	0.51 ± 0.05 ^{ab}	1.18 ± 0.03	1.06 ± 0.11 ^c
	340	0.71 ± 0.01 ^c	0.91 ± 0.13	0.49 ± 0.06 ^{bc}	0.52 ± 0.06 ^{ab}	1.10 ± 0.07	1.12 ± 0.03 ^c
	370	0.45 ± 0.01 ^d	0.83 ± 0.12	0.36 ± 0.01 ^c	0.42 ± 0.03 ^b	0.96 ± 0.05	1.03 ± 0.26 ^c
Carbonate alkalinity	0	0.89 ^p	1.04	0.66 ^p	0.66 ^p	1.29	1.47 ^p
	15	0.64 ^q	0.88	0.42 ^q	0.48 ^q	1.08	1.07 ^q
Protein levels	310	0.96 ^x	1.04	0.53	0.63	1.34	1.20
	340	0.83 ^x	1.00	0.58	0.57	1.20	1.42
	370	0.50 ^y	0.83	0.50	0.52	1.02	1.18
P-value of two-way ANOVA							
Protein levels		0.00	0.20	0.36	0.31	0.22	0.28
Carbonate alkalinity		0.00	0.13	0.00	0.00	0.12	0.00
Carbonate alkalinity * Protein levels		0.03	0.58	0.69	0.73	0.87	0.57

Note: all the values are expressed as the mean values $\pm$ SD, and different letters in the same column denote a significant difference ($p < 0.05$) (the same as those shown in the following tables).

those of fish fed the diets containing the protein level of 310 and 340 g/kg ($p < 0.05$).

3.6. Intestinal Microflora

3.6.1. Microbiota of the intestine richness and diversity analysis

The Alpha diversity statistics of the intestinal microbiota in Songpu mirror carp are presented in Table 11. The Richness estimate of sob and ace, and the Diversity estimators of shannon and simpson were not significantly affected by either the carbonate alkalinity levels or the dietary protein levels ($p > 0.05$). The sob and ace indices in the intestine also tended to decrease with increasing dietary protein levels, although there was no statistically significant difference ($p > 0.05$).

A total of 24 OTUs were shared by the six groups, and the number of unique OTUs in fish fed the diet containing the 370 g/kg protein at a carbonate alkalinity of 15 mmol/L was the highest (Fig. 1). At the same carbonate alkalinity levels, the number of species increased with increasing dietary protein levels from 310 to 370 g/kg and at the same dietary protein levels, the number of species increased with increasing carbonate alkalinity levels from 0 to 15 mmol/L, which was consistent with the change of the sob and ace indices in the intestine.

3.6.2. Microbiota composition, relative abundance analysis and comparison

As shown in Fig. 2, the composition of the intestinal microbiota at the phylum level showed obvious changes. Each treatment group at the phylum level was mainly composed of *Fusobacteria*, *Bacteroidetes* and *Proteobacteria*.

As shown in Table 12., *Fusobacteria* was the most abundant bacterium in all treatment groups. At the same dietary protein levels, the *Proteobacteria* abundance increased with increasing carbonate alkalinity level, although there was no statistically significant difference ($p > 0.05$) and the bacteroidetes levels significantly decreased with increasing carbonate alkalinity level ($p < 0.05$). At the same carbonate alkalinity level, the *Proteobacteria* abundance increased with increasing dietary protein levels, with no significant differences being observed between the groups ($p > 0.05$). The highest *Proteobacteria* abundance was found

Table 11
Microbiota of the gut richness and diversity analysis of Songpu mirror carp under different treatments.

under different treatments					
Carbonate alkalinity levels (mmol/L)	Protein levels(g/kg)	Richness estimate		Diversity estimators	
		sobs	ace	shannon	simpson
0	310	32.0 ± 11.0	44.1 ± 10.0	0.95 ± 0.14	0.49 ± 0.07
		44.1 ± 12.9	52.4 ± 13.2	0.77 ± 0.16	0.61 ± 0.08
	340	42.7 ± 16.2	54.4 ± 12.9	1.05 ± 0.37	0.54 ± 0.11
		24.3 ± 6.4	37.7 ± 4.2	0.72 ± 0.22	0.64 ± 0.13
	370	38.0 ± 7.6	54.1 ± 12.3	0.93 ± 0.11	0.56 ± 0.06
		52.0 ± 10.2	56.8 ± 9.4	1.53 ± 0.44	0.36 ± 0.12
15	310	39.8 ± 38.1	50.3 ± 49.5	0.92 ± 1.06	0.55 ± 0.52
		28.2 ± 41.3	40.9 ± 53.2	0.83 ± 0.85	0.56 ± 0.58
	340	47.3 ± 47.3	55.6 ± 55.6	1.28 ± 1.28	0.45 ± 0.45
	370				
P-value of two-way ANOVA					
Protein levels		0.25	0.37	0.21	0.70
Carbonate alkalinity		0.85	0.94	0.54	0.36
Carbonate alkalinity * Protein levels		0.70	0.90	0.45	0.27

Note: all the values are expressed as the mean values ± SD, and different letters in the same column denote a significant difference ($p < 0.05$) (the same as those shown in the following tables).

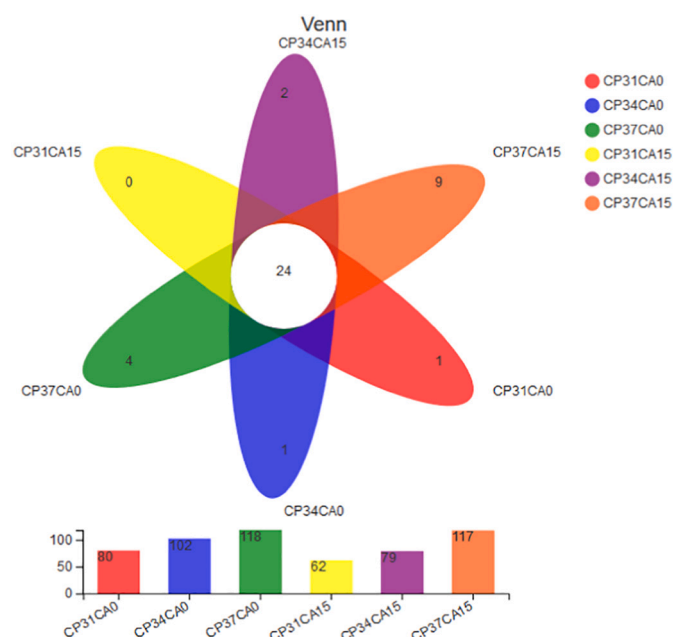


Fig. 1. Venn diagram demonstrating the distribution of OTUs shared by Songpu mirror carp under different treatments.

Note: CP31CA0–310 g/kg crude protein and 0 mmol/L carbonate alkalinity; CP34CA0: 340 g/kg crude protein and 0 mmol/L carbonate alkalinity; CP37CA0: 370 g/kg crude protein and 0 mmol/L carbonate alkalinity; CP31CA15: 310 g/kg crude protein and 15 mmol/L carbonate alkalinity; CP34CA15: 340 g/kg crude protein and 15 mmol/L carbonate alkalinity; CP37CA15: 370 g/kg crude protein and 15 mmol/L carbonate alkalinity. The same as those shown in the following figures).

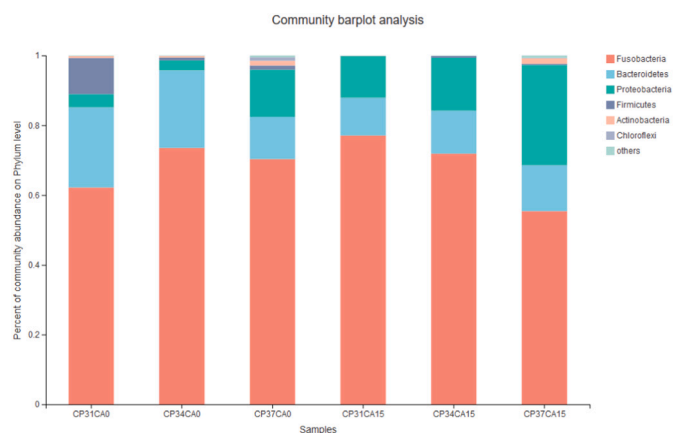


Fig. 2. The bacterial composition of the different communities at the phylum level under different treatment.

in the fish fed the diet containing 370 g/kg protein at a carbonate alkalinity of 15 mmol/L.

Firmicutes abundance and the ratio of *Firmicutes*:*Bacteroidetes* were significantly affected by the carbonate alkalinity and dietary protein, and there was a significant interaction between them ($p < 0.05$). At a carbonate alkalinity of 0 mmol/L, the *Firmicutes* abundance declined with increasing dietary protein levels from 310 to 370 g/kg, whereas at a carbonate alkalinity of 15 mmol/L, few *Firmicutes* were detected in the three treatment groups. At a carbonate alkalinity of 0 mmol/L, the ratio of *Firmicutes*:*Bacteroidetes* significantly decreased with increasing the dietary protein levels from 310 to 370 g/kg ($p < 0.05$).

To explore the changes in the microbial community in depth, a heat map of the relative abundances of the top 50 most abundant bacteria at

Table 12

Relative abundance of *Fusobacteria*, *Bacteroidetes*, *Proteobacteria* and *Firmicutes*, and Ratio of *Firmicutes/Bacteroidetes* in the intestine of Songpu mirror carp under different treatments.

Carbonate alkalinity levels(mmol/L)	Protein levels(g/kg)	<i>Fusobacteria</i>	<i>Bacteroidetes</i>	<i>Proteobacteria</i>	<i>Firmicutes</i>	<i>Firmicutes/Bacteroidetes</i>
0	310	0.62 ± 0.08	0.30 ± 0.04 ^a	0.04 ± 0.01 ^b	0.14 ± 0.03 ^a	0.52 ± 0.18 ^a
	340	0.74 ± 0.08	0.22 ± 0.07 ^{ab}	0.03 ± 0.01 ^b	0.02 ± 0.00 ^b	0.11 ± 0.05 ^b
	370	0.71 ± 0.07	0.15 ± 0.03 ^b	0.13 ± 0.07 ^{ab}	0.01 ± 0.00 ^b	0.06 ± 0.02 ^b
15	310	0.76 ± 0.10	0.18 ± 0.01 ^b	0.12 ± 0.07 ^{ab}	0.00 ± 0.00 ^b	0.00 ± 0.00 ^b
	340	0.73 ± 0.05	0.16 ± 0.02 ^b	0.15 ± 0.03 ^{ab}	0.00 ± 0.00 ^b	0.00 ± 0.00 ^b
	370	0.50 ± 0.13	0.18 ± 0.02 ^b	0.37 ± 0.17 ^a	0.00 ± 0.00 ^b	0.00 ± 0.00 ^b
Carbonate alkalinity	0	0.69	0.22	0.07	0.06 ^p	0.23 ^p
	15	0.66	0.17	0.21	0.00 ^q	0.00 ^q
Protein levels	310	0.69	0.23	0.08	0.07 ^x	0.26 ^x
	340	0.73	0.18	0.09	0.01 ^y	0.05 ^y
	370	0.60	0.16	0.25	0.00 ^y	0.03 ^y
<i>P</i> -value of two-way ANOVA						
Protein levels		0.36	0.17	0.10	0.00	0.02
Carbonate alkalinity		0.71	0.10	0.05	0.00	0.00
Carbonate alkalinity * Protein levels		0.19	0.16	0.64	0.00	0.02

Note: all the values are expressed as the mean values±SD, and different letters in the same column denote a significant difference ($P < 0.05$) (the same as those shown in the following tables).

the genus level is shown in Fig. 3A. At the same dietary protein level, the relative abundances of *Lactococcus*, *Streptococcus* and *Actinomyces* decreased with increasing carbonate alkalinity levels, but *Shewanella*, *vibrio* and *Flavobacterium* obviously increased with increasing carbonate alkalinity levels. At the same carbonate alkalinity levels, increasing dietary protein levels did not significantly affect the microbial community

at the genus level.

Principal coordinate analysis (PCoA) in Fig. 2C also showed that the structure of the intestinal microbiota showed obvious changes with increasing carbonate alkalinity levels. These results indicated that there was a significant structural difference in terms of beta diversity between the two carbonate alkalinity treatment groups. In contrast, there was no

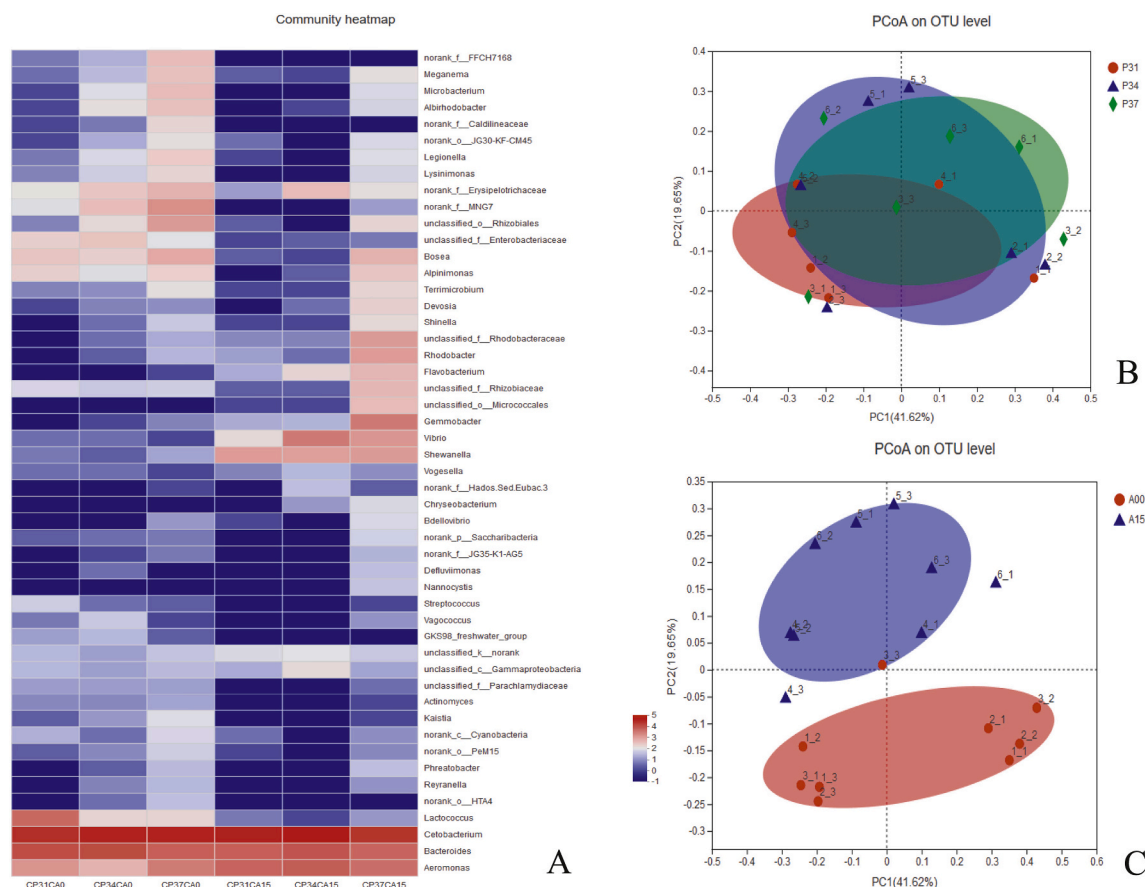


Fig. 3. Changes in the intestinal microbiota at the genus level under different treatments. (A) The heatmaps of the specimens show the relative abundances of the main identified bacteria at the genus taxonomic level. Red indicates a higher relative abundance, whereas blue indicates a lower relative abundance. (B) Principal coordinate analysis (PCoA) of the unweighted UniFrac scores of the microbial communities at the different protein levels. (C) Principal coordinate analysis (PCoA) of the unweighted UniFrac scores of the microbial communities at the different carbonate alkalinity levels. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

significant structural difference in terms of beta diversity among the different dietary protein treatment groups.

4. Discussion

4.1. Carbonate alkalinity effects on growth, digestive ability, intestinal immune, physical barrier function and intestinal microflora

The current study revealed that at a carbonate alkalinity of 15 mmol/L, the growth performance and feed utilization of Songpu mirror carp were restricted. This was shown by a significant decrease in WGR and a significant increase in FCR. The observed changes in the growth performance and feed utilization under the chronic stress of carbonate alkalinity in this study correspond with those found on *Allogynogomphus crucianus*, which showed a significant decrease in the feed ration at a carbonate alkalinity of 10 mmol/L (Lv et al., 2017) and Songpu mirror carp in our laboratory (Fan et al., 2020a, 2020b).

The influence mechanism of carbonate alkalinity on the growth of fish may involve many aspects, among which a decrease in digestion ability is one important factor. For *Lateolabrax japonicus*, significant decreases in the digestive enzyme activities were exhibited when the carbonate alkalinity of the water exceeded 10 mmol/L and with the increase of carbonate alkalinity and the extension of experiment duration, decreases in the digestive enzyme activities were more obvious (Yuan, 2011). This study examined the impact of carbonate alkalinity on digestion ability and found that significantly lower activities of amylase, lipase in the intestine of Songpu mirror carp at a carbonate alkalinity of 15 mmol/L were observed compared to those of fish at a carbonate alkalinity of 0 mmol/L. These results are similar to Cheng et al. (2020), who showed that the activities of protease and lipase in the foregut and midgut were found to be decreased significantly under a carbonate alkalinity stress of 15 mmol/L. As one of the regulatory factors of physiological activities, carbonate alkalinity increases the pH value of the water environment and when it exceeds the optimal upper limit of pH for the digestive enzyme activity in fish, the activity of digestive enzymes declines dramatically, leading to an obvious decline in the digestion ability and feeding levels of fish (Dang et al., 2012).

The intestine also provides an effective combination of immune and physical barrier function (Turner, 2009), especially for fish without a stomach. The antioxidant capacity and activities of the immunologically active material of fish are commonly used as functional indices to investigate immune potential (Xu et al., 2016). A related study has found that during a nine day carbonate alkalinity stress for *Gymnocypris przewalskii*, the activities of both the liver and kidney SOD were initially enhanced by acute carbonate alkalinity stress at 32 and 64 mmol/L with exposure time and then returned to control levels, suggesting that fish can effectively cope with short-term or acute high-concentration carbonate alkaline stress by regulating antioxidative reactions (Wang et al., 2013). Similar results were observed in *Exopalaemon carinicauda* (Liu et al., 2016a, 2016b). In this study, significant reductions in the activities of SOD and T-AOC and significant improvements in the contents of MDA of fish at a carbonate alkalinity of 15 mmol/L were observed after an eight week carbonate alkalinity stress trial. The results demonstrated that the intestinal antioxidant capacity of fish, including neutralization of superoxide anions and surplus free radicals, was weakened by chronic carbonate alkalinity stress, so this variability in the impact of carbonate alkalinity stress among studies on the antioxidant capacity of fish could be related to exposure time. Additionally, this study also found that chronic carbonate alkalinity stress can significantly inhibit the activities of immunologic active materials, such as C3, C4 and Ig M, manifesting that the nonspecific immunity of fish might be suppressed to a certain extent.

Apart from these, the intestinal immune function mainly relies on the cytokines and physical barrier in fish. Cytokines consists of proinflammatory cytokines (TNF- α , IL-1 β , IL-6, IL8, etc.) and anti-inflammatory cytokines (IL-10, TGF- β 2, etc.), which will cause an

immune response (Nakao et al., 2011). Intestinal physical barrier is composed of the epithelial cells and tight junction complex. Tight junctions mainly depends on transmembrane (such as occludin and members of the claudin superfamily) and cytosolic (such as ZO-1) to achieve the purposes of maintaining the epithelial barrier function and controlling the paracellular permeability (Chasiotis et al., 2012). Significant up-regulations in the relative expressions of proinflammatory cytokines (IL-6, IL-8, TNF- α), significant down-regulations in the relative expressions of TJs (occludin, claudin3, claudin7 and MLCK) and a down-regulation in the relative expression of anti-inflammatory (IL-10) were observed in this study. The observed changes in this study, in the cytokines and physical barrier of intestine under stressful conditions are similar to those found by Meng et al. (2018), which reported that tight connection protein (claudin-3, claudin-c) and anti-inflammatory cytokines, signal molecular (TOR) of common carp (*Cyprinus carpio* L) were found down-regulated conversely by high-concentration Cu²⁺ as an acute stresser. For *Ctenopharyngodon idellus* upon acute high-concentration Pb²⁺ stress, the expression levels of inflammation-related genes (IL-1 β , TNF- α and IL-10) and intestinal tight junction proteins (Claudin-3 and ZO-1) were significantly up-regulated (Meng et al., 2018). Two conclusions establishment of a normal or protective microbiota in vivo play a crucial role in excluding potential invaders and maintaining health (Balcazar et al., 2006). The above results revealed that environmental stress conditions such as carbonate alkalinity (in this study), Cu²⁺ (Meng et al., 2018) and Pb²⁺ (Liu et al., 2019a, 2019b) can impair the immune and physical barrier functions, and that different fish have different immune responses mechanism to environmental stress. Thus, those variations on the intestinal immune and physical barrier functions should be used in future studies as a major indice to assess the influence of environmental stress.

Intestinal microflora, considered as a “metabolic organ”, has been widely studied in recent years (Caitriona et al., 2013). The composition and structure of the intestinal microflora of fish, which are the key components in fish health status, are dependent on or conditioned to the immediate environment (Caitriona et al., 2013; Gomez and Balcazar, 2008). In the present study, the intestinal microflora of Songpu mirror carp was perturbed by carbonate alkalinity. In this study, the microbial community of intestine samples of Songpu mirror carp was mainly composed of *Fusobacteriales*, *Bacteroides*, and *Proteobacteria*. Among the microflora species, 60% to 70% of the taxa represented *Fusobacteriales*, almost all whose strains currently detected belong to *Cetobacterium*. For freshwater fish, *Cetobacterium* is a common and widely distributed species in the intestinal tract. The above results are consistent with previous studies of van Kessel et al. (2011) and Meng et al. (2018). Carbonate alkalinity stress had no significant effect on the levels of *Fusobacteriales*, *Bacteroides*, and *Proteobacteria*, whereas increasing the carbonate alkalinity from 0 to 15 mmol/L led to significant reductions in the relative abundance of *Firmicutes* and the *Firmicutes*:*Bacteroidetes* ratio. The *Firmicutes*:*Bacteroidetes* (F/B) ratio is believed to be an important causal factor in development of obesity, increased in obese people in comparison to lean people, and tends to decrease with weight loss (Sweeney and Morton, 2013; Barlow et al., 2015). In this study, a reduction in the relative abundance of *Firmicutes* and the *Firmicutes*:*Bacteroidetes* ratio were positively correlated with reduction in final weight, WGR and CF. Based on these, it was hypothesized that *Firmicutes* might be a more effective and more dominate energy source than bacteroidetes, promoting more efficient absorption of calories and subsequent weight gain for species such as Songpu mirror carp or common carp. Carbonate alkalinity might also directly affect the relative abundance of firmicutes and the *Firmicutes*:*Bacteroidetes* ratio to change the compositions and structure of the intestinal microflora of Songpu mirror carp, resulting in a negative effect on their growth and energy utilization.

Proteobacteria are also important components of the intestine microflora of fish. The present study revealed that most of the *Proteobacteria* found in the intestines of common carp belong to *Aeromonas*, which are a conditional pathogen leading to human-fish-animal

complications, in line with the results of previous studies (Caitriona et al., 2013; Gomez and Balcazar, 2008). The abundances of *Shewanella*, *Vibrio* and *Favobacterium*, which belong to pathogenic bacterium were also observed to increase, whereas *Lactococcus*, *Streptococcus* and *Actinomyces* which belong to nonpathogenic bacterium, were observed to decrease. It appears that high carbonate alkalinity or other environment stress conditions in water could make fish gut become a potentially pathogenic environment and sequentially induce a negative impact on declining the number of pathogenic intestinal microbiota and strengthen the immune function of defend against disease.

4.2. Dietary protein levels effect on growth, intestinal immune and physical barrier function and intestinal microflora

Excessive dietary protein will reduce the growth of juvenile giant grouper (*Epinephelus lanceolatus*) (Gao et al., 2019), tilapia (*Oreochromis niloticus*) (Liu et al., 2017), the Chinese rice field eel (*Monopterus albus*) (Yue et al., 2019) and yellowfin seabream (*Acanthopagrus arabicus*) (Ahmad et al., 2019). The present study revealed that excessive dietary protein response, for Songpu mirror carp, was in line with these authors, with the fish fed diets containing up to 310 g/kg of proteins showing a reduction in both final mean weight and WGR. From the perspective of feed utilization, the values of FCR tended to decrease with increasing dietary protein, which meant that the relatively poor adaptation to the diets could partly result in the lower protein efficiency ratio and the subsequently lower WGR.

The impact of dietary proteins on antioxidant capacity of fish is varied. While some studies suggest a significant effect of dietary proteins (Sun et al., 2016; Xu et al., 2019; Zhang et al., 2016), many other studies are contradictory (Liu et al., 2019a, 2019b; Le et al., 2016). This study found that improvement in content of MDA were observed with increasing dietary proteins, although there was no statistically significant difference among all dietary protein treatments and the activities of SOD and TAOC had no obvious trend. It could be hypothesized that the increment of dietary proteins within the gradient range set in this study both causes oxidative damage for Songpu mirror carp, but also ensures the stability of the antioxidant system and maintains fish health. Although the antioxidant system was unaffected, it was found that dietary protein levels up to 310 g/kg significantly decreased both the C3 and C4 contents in the hepatopancreas and the relative expressions of IL-6 and TNF- α in intestine to weaken the immune function and down-regulated the relative expressions of TJs and MLCK in intestine to destroy the intestinal intercellular structural integrity. The possible reason for this is that in comparison to the indicators related to antioxidant capacity, immunologic active material C3 and C4, cytokines and TJs, which are related to immune and physical barrier function are more sensitive to protein nutrition for Songpu mirror carp. A contrary pattern of responses has been documented in other fish species, such as Jinbian carp (Le et al., 2016), crucian carp (Cai et al., 2001), koi (Jiang et al., 2012), *Barbus capito* (Wang et al., 2014). This discrepancy is probably due to the differences in fish strains, environmental conditions, or diet composition used in these studies.

As a crucial nutrient element in fish diets, many studies suggest that dietary protein sources exert a noticeable impact on the intestine microflora of fish. A study on Atlantic cod indicated that when fish meal served as dietary protein source, the dominant bacteria of gram-positive bacteria in intestine mainly belong to cycloseria and carnivorous bacilli, but when fish meal was replaced with soybean meal, the dominant bacteria in intestine mainly belong to helicobacter pylori (Ringo et al., 2006). For rainbow trout, the number of aerobic bacteria and anaerobic bacteria in intestine tended to decrease when partly replacing fish meal with soybean meal (Heikkinen et al., 2006). Apart from protein sources, protein addition levels in diets also play a delicate role in the intestine microflora of fish. Research on *Trachinotus ovatus* found that the clustering analysis indicated that bacteria communities of different intestinal segments showed furthest distances which meant diets containing

44% protein influenced the intestinal bacteria community structure of *Trachinotus ovatus* the most, but the sequencing results of bands cut from denaturing gradient gel electrophoresis (DGGE) gels showed that the dominant bacteria of *Trachinotus ovatus* intestine mainly belonged to *Proteobacteria*, *Firmicutes* and *Chloroflexi* (Tang, 2014). This study also found that increasing dietary protein levels had no effect on the diversity of intestine microflora and the dominant bacteria of Songpu mirror carp intestine mainly belonged to *Fusobacteria*, *Bacteroidetes* and *Proteobacteria*, suggesting that the dietary protein level could change the structure of the intestine microflora to some extent, but the dominant bacteria were not affected. From the perspective of microflora composition, the *Proteobacteria* abundance increased significantly with increasing dietary protein, but the *Firmicutes* abundance and *Firmicutes*:*Bacteroidetes* ratio, which encode a wide range of carbohydrate-related enzymes and play an irreplaceable role in the utilization of non-digestible polysaccharides (Mariat et al., 2009), were reduced. This implied the growth depression caused by high-concentration dietary proteins might be attributed to the hypothesis that fish fed with excessive dietary protein would use additional energy to handle increased deamination (Wilson, 2020; Gan et al., 2012), resulting in insufficient nonprotein energy available for growth by decreasing the *Firmicutes* abundance and *Firmicutes*:*Bacteroidetes* ratio.

4.3. The interaction between carbonate alkalinity and dietary protein with overtime

Studies on *Oreochromis niloticus* revealed that dietary protein level, which reached or exceeded 31.04%, would effectively improve anti-stress and anti-oxidation capabilities, thereby enhancing the body's resistance to ammonia-nitrogen stress (Wang, 2018). In a previous study in our laboratory, it was shown that a 10 g AKG/kg supplementation in diets of Songpu mirror carp could enhance the immunity, relieve the immune stress and strengthen the ability to resist chronic carbonate alkalinity stress (Fan et al., 2020a, 2020b). These findings imply that nutrition regulation can achieve the purpose of improving the immune function of fish and relieving the stress of external environment to a certain extent. In the present study, it was hypothesized that the impact of carbonate alkalinity on growth, digestive ability, intestinal immune, physical barrier function and intestinal microflora might be enhanced by increasing dietary proteins and was confirmed that as far as it concerns the C3 content, the relative expression of occludin and the abundance of firmicutes and *Firmicutes*:*Bacteroidetes* ratio, there was an interaction between carbonate alkalinity and dietary proteins over time. At the high-concentration carbonate alkalinity stress of 15 mmol/L, increasing dietary proteins from 310 g/kg to 370 g/kg significantly reduced the above indexes; similarly, at the same dietary protein treatment, increasing carbonate alkalinity also significantly reduced the above indexes and led to lower final weight and WGR. At the same dietary protein treatment, increasing carbonate alkalinity also resulted in lower final weight and WG, although there was no significant difference among all treatments, so under carbonate alkalinity stress, excessive dietary proteins would not promote the growth and elevate the immune function of fish.

5. Conclusions

Carbonate alkalinity significantly impacted the growth, digestive ability, intestinal immune, physical barrier function, and especially on intestinal microflora. Simultaneously, at a carbonate alkalinity of 15 mmol/L, increasing dietary protein levels from 310 g/kg to 370 g/kg had an adverse effect on the growth performance and immune function, and brought about the anomalous change of intestinal microflora. Therefore, under the long-term carbonate alkalinity stress, dietary protein level for Songpu mirror carp should be controlled within appropriate limits, which could effectively weaken the oxidative damage produced by carbonate alkalinity stress and further strengthen the

growth and intestinal immune response of fish.

Data availability statement

All data generated or used during the study appear in the submitted article.

Declaration of Competing Interest

The authors declare that there are no conflicts of interest.

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